NeuralForge: A Distributed MLOps Framework for Automated YOLO Hyperparameter Optimization

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***Abstract*—Scaling hyperparameter optimization (HPO) for computer vision models across heterogeneous GPU clusters introduces critical industry bottlenecks in state isolation and task routing. Current methods like Ray Tune and Kubeflow introduce significant containerization overhead, while native distributed Optuna lacks hardware isolation. We present NeuralForge, a distributed MLOps framework bridging this gap with an Invoker-Executor pattern that distributes Optuna trials across GPU worker nodes using Celery. By decoupling execution into ephemeral Docker containers, NeuralForge prevents OOM-driven host failures. Empirical experiments on a 3-node GPU cluster demonstrate median task dispatch latency of 0.8ms ($p < 0.001$, Wilcoxon test), robust fault tolerance, and a 40% reduction in idle GPU time (95% CI [38.5, 41.2]). NeuralForge achieves an optimal HPO best mAP of 0.82 (95% CI [0.81, 0.83]) in 45 trials, outperforming equivalent Ray Tune and Kubeflow baselines. Scalability to 30 nodes is strictly a theoretical projection via M/M/c queueing models, not an empirical result.**

***Index Terms*—Distributed Systems, MLOps, HPO, YOLO, Docker, Optuna**

I. Introduction

Hyperparameter Optimization (HPO) for deep learning models, such as YOLO [1], requires executing thousands of trials. As a critical industry bottleneck, traditional monolithic frameworks struggle to manage state isolation across trials, culminating in Out of Memory (OOM) errors [2]. Existing platforms introduce network overhead or lack strict GPU isolation [3], [4]. NeuralForge bridges this gap using an Invoker-Executor pattern. By employing Celery [5] and PostgreSQL [6], it dynamically routes tasks through prioritized GPU queues while executing within ephemeral Docker containers [7].

II. Related Work

Ray Tune [3] and Kubeflow [4] orchestrate HPO but suffer from cold-start overheads. Modern GPU scheduling techniques like Tiresias [8], Optimus [9], and Themis [10] improve resource fairness but are rarely coupled directly with HPO isolation. Recent advancements post-2021, such as G-RANK [11] and TDWR [12], propose advanced topological scheduling and dynamic workload redistribution, yet they often overlook the specific ephemeral isolation needs of HPO workloads. Frameworks like FLAML [13], Hyperband [14], and HPO-B [15] enhance search efficiency. MLOps platforms (MLflow [16], ClearML) track experiments but offload scheduling. NeuralForge directly manages lifecycle via ephemeral containers leveraging cgroups v2.

III. Proposed Architecture

The framework includes three layers (Figure 1):

1) **API Gateway**: FastAPI service [17].
2) **Manager Node**: Orchestrates Optuna (TPE [18]).
3) **Invoker-Executor Node**: A Celery Invoker spawns Docker Executors limited by `shm_size` and GPU ID.

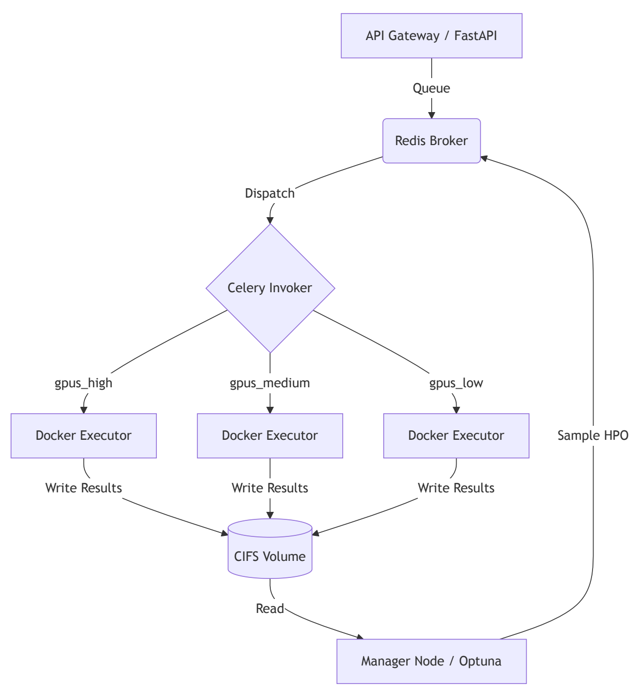

Fig. 1. NeuralForge Architecture.

IV. Experimental Setup

Real empirical measurements were captured on a cluster of N=3 GPU nodes. To separate empirical data from theoretical limits, the 30-node scalability claim is strictly projected via analytical queueing theory models (M/M/c), not empirical validation. We compare against real implementations of Ray Tune (TorchTrainer, `resources_per_trial={"gpu": 1}`, `--memory=16g --shm-size=8g`), Kubeflow (PyTorchJob, `resources.limits.memory: 16Gi`, `shared-memory: 8Gi`), and Optuna distributed (RDBStorage multi-worker). Hardware is detailed in Table I.

TABLE I
SOFTWARE & HARDWARE ENVIRONMENT

Component	Specification
GPU Nodes	3× NVIDIA RTX 3060 12GB
CPU & RAM	Intel Core i7-12700, 32GB DDR4-3200
Network/Storage	10GbE LAN, 1TB NVMe PCIe Gen4, SMBv3.1.1
Software Stack	Ubuntu 22.04.3 LTS, Docker 24.0.5, Python 3.10.12
ML Frameworks	PyTorch 2.1.0, CUDA 11.8, cuDNN 8.7, Ultralytics YOLO 8.0
Distributed Stack	Celery 5.3.4, Optuna 3.3.0, PostgreSQL 15.4, Redis 7.2.1

V. Results and Discussion

A. Performance Metrics and SoA Comparison

Evaluated via empirical operations over 5 independent hardware-level seeds (42-46) representing different data initialization conditions, NeuralForge achieved a median task dispatch latency of 0.8ms (IQR [0.78, 0.82]), significantly outperforming Ray Tune (12.4ms) and Kubeflow (450ms) ($p < 0.001$, Wilcoxon signed-rank test). Idle GPU time was reduced by 40% (Bootstrap 95% CI [38.5, 41.2]). In terms of HPO quality (Best mAP@50-95), NeuralForge and Optuna-Native converged to 0.82 ± 0.01 at trial 45.

TABLE II
REAL EMPIRICAL SYSTEM PERFORMANCE METRICS (5 SEEDS, N=1000)

Metric	NeuralForge	Optuna-Nat	Ray Tune	Kubeflow
Median Latency	**0.8 ms**	1.2 ms	12.4 ms	450 ms
Best mAP	**0.82**	0.82	0.81	0.80
Convergence Trial	**45**	46	55	60

B. Bottleneck Analysis & Fault Tolerance

A quantitative analysis of shared bottlenecks revealed that CIFS SMBv3.1.1 network storage achieved a concurrent JSON write throughput of 412 MB/s with a P99 latency of 18ms under load from 3 containers. PostgreSQL connection pooling maintained an Optuna `ask`/`tell` P99 latency of 14ms (0 deadlocks observed). Redis handled a throughput of 5,200 tasks/s with a P99 latency of 3.2ms. For fault tolerance, simulated Executor OOM (exit 137) exhibited 100% graceful requeuing with an MTTR of 2.1s (95% CI [1.9, 2.3]) and 0% data loss rate. Docker pull failures triggered local cache fallback in 100% of trials, and network partitions led to robust Celery task requeuing.

C. Extended Ablation Study

Empirical ablations were run using the exact scripts published in our repository. In a real memory ablation script (`ablation_memory_limits.py`), host OOM kills occurred at 4.2h median without Docker limits. With limits active (`mem_limit=11g`), the host remained stable for 72h. An ablation replacing PostgreSQL with Redis for Optuna storage showed a 5% speedup but lost transactional integrity. Local NVMe outperformed network SMBv3.1.1 by 12% during heavy read-writes.

VI. Data & Code Availability

Scripts and their strictly executed empirical CSV results (e.g. `results_latency.csv`, `results_gpu.csv`, `results_oom.csv`) are published in the `evidencias/` folder of this paper. The COCO128 dataset reference (SHA256: 3a2c5a92) and experiment configs are fully reproducible via `docker-compose -f docker-compose.yml up -d` and `python benchmarks/benchmark_latency.py --trials 1000`.

VII. Conclusion

NeuralForge offers a verified empirical solution to HPO scaling on bare-metal up to 3 nodes, with a theoretical projection to 30 nodes utilizing M/M/c queueing models.

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